A real life example to explain data, information, insights

- Data is the raw numbers that we capture according to some agreed to standards. Having consistent standards is very important since having data recorded according to different standards can be extremely problematic. For example, there is the age old question of how long is a piece of string? The answer depends on what measurement standard you are using. If we use the Metric system we may come up with some number. That number can vary depending on if we are using meters, centimeters, or millimeters. Using British Imperial/US Customary units will result in an entirely different number. As such, one of the most important steps in any analytics effort is defining standards we are applying. When doing analytics projects, one of our first tasks is to go through the client's current data structure and normalize that data. In other words, we make sure that all the like things are being measured in the same way.

- Information is a collection of data points that we can use to understand something about the thing being measured. Going back to our string example, let’s say we have 100 pieces of string that have been produced by our company. We have agreed to measure them in centimeters and to record the length of each of those pieces of string. Each measurement is considered a data point. Taken together, however, all these data points provide some very useful information. For example, if we need all the pieces of string to be 10 cm long, but many of them are not, we then know that something is not quite right with our process and we might need to take some action to address this issue. How do we know what actions to take? For that we need insights.

What Does Insight Mean?

- Insight is gained by analyzing data and information to understand what is going on with the particular situation or phenomena. Finding the insight can then be used to make better business decisions. Going back to our pieces of string, if we know that our pieces of string are not what we want, we now have to analyze the data to determine what actions to take. So for example, if we find that our mean (i.e., average) length of string is 9.5 cm we now know that our strings tend to be shorter than we want. To make the right business decisions, we need to know how consistently our process produces these pieces of string. To find that out we look at how much variance there is in the data, as well as the upper and lower limits of our outputs. One approach is to look at the standard deviation, which is a statistic that tells us the average amount by which the various measures deviate from the mean. If we found in this case that our standard deviation was say .25 cm we know that even though the strings are too short, they are fairly consistent. Our insight might be that while our process is not what we want it to be it is doing things the same way most of the time.

Finding Insights in Data

- Now we can look at our upper and lower limits to see what our longest and shortest piece of strings are. Let’s say we find that our longest piece of string is only 9.75 cm and our shortest is only 9.35 cm, this would mean that there is a pretty good chance that our process for producing the pieces of string are simply cutting them too short. This insight is a finding that we need to check our machine and adjust where and when it cuts. Since not all the pieces of string are coming out the same size, however, we know that there are other issues that may be driving out process not being what we want. To address those we need to gather more data after we make our change, compile that data into new information, the analyze it to find new insights until we can make the necessary business decisions to drive our desired outcomes.

Importance of storytelling

- Why data alone fails -
- According to the Bureau of Labor Statistics, the demand for research analysts is expected to grow 25 percent between 2020 and 2030, much faster than the average across all industries. Many companies have begun including data storytelling as a required skill in analyst job descriptions, while others have opted to hire for data storyteller positions to supplement their existing analytics teams' abilities. Possessing the skills to both analyze data and communicate its insights can help you stand out as a well-rounded candidate.

Components

- There are three key components to data storytelling:
- Data: Thorough analysis of accurate, complete data serves as the foundation of your data story. Analyzing data using descriptive, diagnostic, predictive, and prescriptive analysis can enable you to understand its full picture.
- Narrative: A verbal or written narrative, also called a storyline, is used to communicate insights gleaned from data, the context surrounding it, and actions you recommend and aim to inspire in your audience.
- Visualizations: Visual representations of your data and narrative can be useful for communicating its story clearly and memorably. These can be charts, graphs, diagrams, pictures, or videos.

The Psychological Power of Storytelling

- Humans have told stories since the Cro-Magnon era to communicate with others for survival and record accounts of daily life. While storytelling methods have come a long way since the days of cave paintings, its psychological power holds true tens of thousands of years later.
- The brain's preference for stories over pure data stems from the fact that it takes in so much information every day and needs to determine what's important to process and remember and what can be discarded.

When someone hears a story, multiple parts of the brain are engaged, including:

- Wernicke’s area, which controls language comprehension
- The amygdala, which processes emotional response
- Mirror neurons, which play a role in empathizing with others

- Rather than presenting your team with a spreadsheet of data and rattling off numbers, consider how you can engage multiple parts of their brains. Using data storytelling, you can evoke an emotional response on a neural level that can help your points be remembered and acted upon.

How to Craft a Compelling Data Narrative

- Data storytelling uses the same narrative elements as any story you've read or heard before: characters, setting, conflict, and resolution.
- To help illustrate this, imagine you're a data analyst and just discovered your company's recent decline in sales has been driven by customers of all genders between the ages of 14 and 23. You find that the drop was caused by a viral social media post highlighting your company's negative impact on the environment, and craft a narrative using the four key story elements:

- Setting: Set the scene by explaining there's been a recent drop in sales driven by customers of all genders, ages 14 to 23. Use a data visualization to show the decline across audience types and highlight the largest drop in young users.
- Conflict: Describe the root issue: A viral social media post highlighted your company's negative impact on the environment and caused tens of thousands of young customers to stop using your product. Incorporate research (such as this article in the Harvard Business Review) about how consumers are more environmentally conscious than ever and how sustainably-marketed products can potentially drive more revenue than their unsustainable counterparts. Remind the team of your company's current unsustainable manufacturing practices to clarify why customers stopped purchasing your product. Use visualizations here, too.

- Resolution: Propose your solution. Based on this data, you present a long-term goal to pivot to sustainable manufacturing practices. You also center marketing and public relations efforts on making this pivot visible across all audience segments. Use visualizations that show the investment required for sustainable manufacturing practices can pay off in the form of earning customers from the growing environmentally conscious market segment.

- If there isn’t a conflict in your data story—for instance, if the data showed your current marketing campaign was driving traffic and exceeding your goal—you can skip that element and go straight to recommending that the current course of action be maintained.
- Whatever story the data tells, you can communicate it effectively by formatting your narrative with these elements and walking your audience through each piece with the help of visualizations.

Communicating the Need for Action

- Data storytelling can help turn data insights into action. Without effective communication, insights can go unnoticed or unremembered by your audience; both hard and soft skills are crucial for leveraging data to its fullest potential.
- Harvard Business School Professor Jan Hammond speaks to this in the online course Business Analytics, one of the courses that make up the Core Business Essentials program.
- "Always remember that applying analytical techniques to managerial problems requires both art and science," Hammond says. "Over my career, I've learned that it's the soft skills that are the hardest to master, but they're critically important."